

2 RELATED WORK

LLM Agents. LLM-based agents treat large language models as the “brain” that plans, acts, and reflects while interfacing with external tools and environments (Wang et al., 2024; Guo et al., 2024). Early formulations follow a think-act loop: ReAct interleaves chain-of-thought with grounded actions (Yao et al., 2023b; Wei et al., 2022), and Reflexion adds self-critique and memory to correct future behavior (Shinn et al., 2023). Building on this, tool-augmented frameworks formalize function calling, tool selection, and state management to operate retrieval, code execution, and other systems (Schick et al., 2023; Lewis et al., 2020). These designs have made LLM agents effective and flexible across domains (He et al., 2024b; Trivedi et al., 2022; Li et al., 2023b). Web agents perceive pages and perform UI-level actions for tasks like shopping and booking (Yao et al., 2022; Zhou et al., 2023; He et al., 2024a; Zheng et al., 2024); code agents iteratively edit, test, and debug multi-file projects (Yang et al., 2024; Hong et al., 2024); embodied agents couple language with perception and control (Kim et al., 2024; Mao et al., 2023; Zhang et al., 2024a); and scientific agents use tools and simulation for literature review, hypothesis generation, and experiment planning (Park et al., 2023; Ren et al., 2025).

Tool-use in LLM agents. Tool use is a core capability of LLM agents and has drawn substantial interest (Qu et al., 2025; Qin et al., 2024). Methods advance when and how models invoke functions: RL approaches (ReTool, ToolRL, OTC) optimize selection and calling (Feng et al., 2025; Qian et al., 2025a; Wang et al., 2025); supervised/instruction tuning (e.g., Toolformer; small-model instruction tuning) teaches adherence to schemas (Schick et al., 2023; Shen et al., 2024); and feedback frameworks (TRICE) add self-correction (Qiao et al., 2023). Complementary evaluations measure these abilities: Gorilla grounds calls in public APIs (Patil et al., 2024); the Berkeley Function-Calling Leaderboard scores cross-domain execution (Patil et al.); ToolBench scales to multi-step RapidAPI tools (Qin et al., 2023); ToolQA targets tool-augmented reasoning (Zhuang et al., 2023); and Meta-Tool probes tool awareness and selection (Huang et al., 2023). Together, this work advances both capability and measurement for tool-using agents. However, existing benchmarks seldom treat tool use as the primary objective. Most prioritize end-task scores and overlook trajectory-level signals, limiting the insight into agent behavior and opportunities for improvement. Therefore, a benchmark with a high-fidelity, diverse tool set and realistic, task-driven queries spanning difficulty levels, enabling trajectory-aware evaluation and diagnosis, is still urgently needed.